EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$ then one of the values of 'k' is -

(A) 1/8 (B) 8 (C) 4 (D) 1/4

MPR143

2. A variable circle is drawn to touch the line $3x - 4y = 10$ and also the circle $x^2 + y^2 = 1$ externally then the locus of its centre is -

(A) straight line (B) circle

(C) pair of real, distinct straight lines (D) parabola

MPR144

3. The straight line $y = m(x - a)$ will meet the parabola $y^2 = 4ax$ in two distinct real points if

(A) $m \in R$ (B) $m \in [-1, 1]$

(C) $m \in (-\infty, 1] \cup [1, \infty)$ (D) $m \in R - \{0\}$

MPR145

4. If on a given base, a triangle be described such that the sum of the tangents of the base angles is a constant, then the locus of the vertex is:

(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola

MPR146

5. The equation of the circle drawn with the focus of the parabola $(x - 1)^2 - 8y = 0$ as its centre and touching the parabola at its vertex is :

Rankers Academy JEE

(A) $x^2 + y^2 - 4y = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$

(C) $x^2 + y^2 - 2x - 4y = 0$ (D) $x^2 + y^2 - 2x - 4y + 1 = 0$

MPR147

6. If a focal chord of $y^2 = 4x$ makes an angle $\alpha, \alpha \in \left(0, \frac{\pi}{4}\right]$ with the positive direction of x-axis, then minimum length of this focal chord is -

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) 8 (D) 16

MPR148

7. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha,0)$ $(\beta,0)$ both to the right of the origin. A circle also passes through these two points. The length of a tangent from the origin to the circle is

(A) $\sqrt{\frac{bc}{a}}$ (B) ac^2 (C) $\frac{b}{a}$ (D) $\sqrt{\frac{c}{a}}$

MPR149

8. The length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c, then

(A) $2a^2 = bc$ (B) $a^3 = b^2c$ (C) $ac = b^2$ (D) $b^2c = 4a^3$

MPR150

9. Locus of trisection point of any arbitrary double ordinate of the parabola $x^2 = 4by$, is -

(A) $9x^2 = by$ (B) $3x^2 = 2by$ (C) $9x^2 = 4by$ (D) $9x^2 = 2by$

MPR151

For More Join: @IITJEE_Advanced

10. Consider the graphs of $y = Ax^2$ and $y^2 + 3 = x^2 + 4y$, where A is a positive constant and $x, y \in R$. Number of points in which the two graphs intersect, is-

(A) exactly 4

(B) exactly 2

(C) at least 2 but the number of points varies for different positive values of A.

(D) zero for atleast one positive A.

MPR152

11. From an external point P, pair of tangent lines are drawn to the parabola, $y^2 = 4x$. If θ_1 & θ_2 are the inclinations of these tangents with the axis of x such that, $\theta_1 + \theta_2 = \frac{\pi}{4}$, then the locus of P is:

(A) $x - y + 1 = 0$ (B) $x + y - 1 = 0$

(C) $x - y - 1 = 0$ (D) $x + y + 1 = 0$

MPR153

12. y-intercept of the common tangent to the parabola $y^2 = 32x$ and $x^2 = 108y$ is

(A) – 18 (B) – 12 (C) – 9 (D) – 6

MPR154

13. The points of contact Q and R of tangent from the point $P(2, 3)$ on the parabola $y^2 = 4x$ are

(A) (9, 6) and (1, 2) (B) (1, 2) and (4, 4)

(C) (4, 4) and (9, 6) (D) (9, 6) and $(\frac{1}{4}, 1)$

MPR155

Rankers Academy JEE

14. If the lines $(y - b) = m_1(x + a)$ and $(y - b) = m_2(x + a)$ are the tangents to the parabola $y^2 = 4ax$, then

(A) $m_1 + m_2 = 0$ (B) $m_1m_2 = 1$

(C) $m_1m_2 = -1$ (D) $m_1 + m_2 = 1$

MPR156

15. The equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is -

(A) $\sqrt{3}y = 3x + 1$ (B) $\sqrt{3}y = -(x + 3)$

(C) $\sqrt{3}y = x + 3$ (D) $\sqrt{3}y = -(3x + 1)$

MPR157

16. Equation of the other normal to the parabola $y^2 = 4x$ which passes through the intersection of those at (4,–4) and (9, –6) is -

(A) $5x - y + 115 = 0$ (B) $5x + y - 135 = 0$

(C) $5x - y - 115 = 0$ (D) $5x + y + 115 = 0$

MPR158

17. Suppose that three points on the parabola $y = x^2$ have the property that their normal lines intersect at a common point (a, b). The sum of their x-coordinates is -

(A) 0 (B) $\frac{2b-1}{2}$ (C) $\frac{a}{2}$ (D) $a + b$

MPR159

For More Join: @IITJEE_Advanced

18. The normal chord of a parabola $y^2 = 4ax$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is:

(A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$

MPR160

19. Which one of the following lines cannot be the normals to $x^2 = 4y$?

(A) $x - y + 3 = 0$ (B) $x + y - 3 = 0$

(C) $x - 2y + 12 = 0$ (D) $x + 2y + 12 = 0$

MPR161

20. Tangents are drawn from the points on the line $x - y + 3 = 0$ to parabola $y^2 = 8x$. Then the variable chords of contact pass through a fixed point whose coordinates are:

(A) (3,2) (B) (2,4) (C) (3,4) (D) (4,1)

MPR162

21. Consider two curves $C_1 : \left(y-\sqrt{3}\right)^2 = 4\left(x-\sqrt{2}\right)$ and $C_2 : x^2 + y^2 = \left(6+2\sqrt{2}\right)x + 2\sqrt{3}y - 6\left(1+\sqrt{2}\right)$, then-

(A) C_1 and C_2 touch each other only at one point.

(B) C_1 and C_2 touch each other exactly at two points.

(C) C_1 and C_2 intersect (but do not touch) at exactly two points.

(D) C_1 and C_2 neither intersect nor touch each other.

MPR163

22. C is the centre of the circle with centre (0, 1) and radius unity. P is parabola $y = ax^2$. The set of values of 'a' for which they meet at a point other than the origin, is-

Rankers Academy JEE

(A) $a > 0$ (B) $a \in \left(0, \frac{1}{2}\right)$ (C) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \infty\right)$

MPR164

23. If the locus of the middle points of the chords of the parabola $y^2 = 2x$ which touches the circle $x^2 + y^2 - 2x - 4 = 0$ is given by $(y^2 + 1 - x)^2 = \lambda(1 + y^2)$, then the value of λ is equal to-

(A) 3 (B) 4 (C) 5 (D) 6

MPR165

24. PN is an ordinate of the parabola $y^2 = 4ax$ (P on $y^2 = 4ax$ and N on x-axis). A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q. NQ meets the tangent at the vertex in a point T such that $AT = kNP$, then the value of k is (where A is the vertex)

(A) 3/2 (B) 2/3 (C) 1 (D) none

MPR166

25. Let A and B be two points on a parabola $y^2 = x$ with vertex V such that VA is perpendicular to VB and θ is the angle between the chord VA and the axis of the parabola. The value of $\frac{|VA|}{|VB|}$ is

(A) $\tan\theta$ (B) $\tan^3\theta$ (C) $\cot^2\theta$ (D) $\cot^3\theta$

MPR167

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 8x$ is a parabola whose-

(A) Latus rectum is half the latus rectum of the original parabola

(B) Vertex is (1,0)

(C) Directrix is y-axis

(D) Focus has the co-ordinates (2,0)

MPR168

2. Consider a circle with its centre lying on the focus of the parabola, $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle & the parabola is

(A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -p\right)$ (C) $\left(-\frac{p}{2}, p\right)$ (D) $\left(-\frac{p}{2}, -p\right)$

MPR169

3. P is a point on the parabola $y^2 = 4ax (a > 0)$ whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x-axis at a point whose co-ordinates are :

(A) $(-3a,0)$ (B) $(-a,0)$ (C) $(-2a,0)$ (D) $(a,0)$

Rankers Academy JEE

MPR170

4. If from the vertex of a parabola $y^2 = 4x$ a pair of chords be drawn at right angles to one another and with these chords as adjacent sides a rectangle be made, then the locus of the further end of the rectangle is -

(A) an equal parabola

(B) a parabola with focus at (9,0)

(C) a parabola with directrix as $x - 7 = 0$

(D) a parabola having tangent at its vertex $x = 8$

MPR171

5. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to axis of x at G, then locus of G is a parabola with -

(A) length of latus rectum equal to $4a$

(B) vertex at $(4a,0)$

(C) directrix as the line $x - 3a = 0$

(D) focus at $(5a,0)$

MPR172

6. TP and TQ are tangents to parabola $y^2 = 4x$ and normals at P and Q intersect at a point R on the curve. The locus of the centre of the circle circumscribing ΔTPQ is a parabola whose

(A) vertex is (1, 0)

(B) foot of directrix is $\left(\frac{7}{8}, 0\right)$

(C) length of latus-rectum is $\frac{1}{4}$.

(D) focus is $\left(\frac{9}{8}, 0\right)$.

MPR173

For More Join: @IITJEE_Advanced

7. The locus of the point of intersection of those normals to the parabola $x^2 = 8y$ which are at right angles to each other, is a parabola. Which of the following hold(s) good in respect of the locus?

(A) Length of the latus rectum is 2.

(B) Coordinates of focus are $\left(0, \frac{11}{2}\right)$

(C) Equation of a director circle is $2y - 11 = 0$

(D) Equation of axis of symmetry is $y = 0$

MPR174

8. Let A be the vertex and L the length of the latus rectum of the parabola, $y^2 - 2y - 4x - 7 = 0$. The equation of the parabola with A as vertex, $2L$ the length of the latus rectum and the axis at right angles to that of the given curve is:

(A) $x^2 + 4x + 8y + 4 = 0$ (B) $x^2 + 4x - 8y + 12 = 0$

(C) $x^2 + 4x + 8y + 12 = 0$ (D) $x^2 + 8x - 4y + 8 = 0$

MPR175

9. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 4ax$ is a parabola whose

(A) Latus rectum is half the latus rectum of the original parabola

(B) Vertex is $(a/2, 0)$

(C) Directrix is y-axis

(D) Focus has the co-ordinates $(a, 0)$

MPR176

10. Let $y^2 = 4ax$ be a parabola and $x^2 + y^2 + 2bx = 0$ be a circle. If parabola and circle touch each other externally then:

(A) $a > 0, b > 0$ (B) $a > 0, b < 0$ (C) $a < 0, b > 0$ (D) $a < 0, b < 0$

Pankers Academy JEE

MPR177

11. P is a point on the parabola $y^2 = 4x$ where abscissa and ordinate are equal. Equation of a circle passing through the focus and touching the parabola at P is:

(A) $x^2 + y^2 - 13x + 2y + 12 = 0$ (B) $x^2 + y^2 - 3x - 18y + 2 = 0$

(C) $x^2 + y^2 + 13x - 2y - 14 = 0$ (D) $x^2 + y^2 - x = 0$

MPR178

12. Subset of complete set of values of m for which a chord of slope m of the circle $x^2 + y^2 = 4$ touches parabola $y^2 = 4x$, can be

(A) $\left(-\infty, -\sqrt{\frac{\sqrt{2}-1}{2}}\right)$ (B) $(0, 1/2)$ (C) $\left(\sqrt{\frac{\sqrt{2}-1}{2}}, \infty\right)$ (D) $(-1/2, 0)$

MPR179

13. Locus of the centre of the circle passing through the vertex and the mid-points of perpendicular chords from the vertex of the parabola $y^2 = 4ax$ is

(A) is a parabola with vertex $(-a, a)$ (B) is a parabola with latus rectum a

(C) is a parabola with vertex $(2a, 0)$ (D) is a parabola with latus rectum $\frac{a}{2}$

MPR180

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 14 to 16

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R. Both the circles C_1 and C_2 pass through the focus of the parabola.

14. Area of the ΔPQR equals

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) $\frac{1}{4}$

MPR181

15. Radius of the circle C_2 is

(A) $5\sqrt{5}$ (B) $5\sqrt{10}$ (C) $10\sqrt{2}$ (D) $\sqrt{210}$

MPR182

16. The common chord of the circles C_1 and C_2 passes through the

(A) incentre of the ΔPQR (B) circumcenter of the ΔPQR

(C) centroid of the ΔPQR (D) orthocenter of the ΔPQR

MPR183

Paragraph for Question No. 17 to 18

The normal at any point P to the parabola $y^2 = 8x$ makes an angle $\theta \in \left(0, \frac{\pi}{4}\right]$ with the positive direction of x-axis.

On the basis of above information, answer the following questions:

17. Maximum distance of the point P from origin is $2\sqrt{\lambda}$, then λ is-

MPR184

18. Maximum focal distance of P is –

MPR185

Rankers Academy JEE

MATCHING LIST TYPE QUESTION

19. Consider the parabola $y^2 = 12x$

	List-I		**List-II**
(I)	Tangent and normal at the extremities of the latus rectum intersect the x axis at T and G respectively. The coordinates of the middle point of T and G are	(P)	(0, 0)
(II)	Variable chords of the parabola passing through a fixed point K on the axis, such that sum of the squares of the reciprocals of the two parts of the chords through K, is a constant. The coordinate of the point K are	(Q)	(3, 0)
(III)	All variable chords of the parabola subtending a right angle at the origin are concurrent at the point	(R)	(6, 0)
(IV)	AB and CD are the chords of the parabola which intersect at a point E on the axis. The radical axis of the two circles described on AB and CD as diameter always passes through	(S)	(12, 0)

(A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ S

(B) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P

(C) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ P

(D) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ S

MPR186

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

20. AB is a chord of the parabola $y^2 = 4ax$ joining $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$. Match the following

	Column - I		**Column - II**
(A)	AB is a normal chord	(P)	$t_2 = -t_1 + 2$
(B)	AB is a focal chord	(Q)	$t_2 = -\frac{4}{t_1}$
(C)	AB subtends 90° at (0, 0)	(R)	$t_2 = -\frac{1}{t_1}$
(D)	AB is inclined at 45° to the axis of parabola	(S)	$t_2 = -t_1 - \frac{2}{t_1}$

MPR187

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. If $(a^2, a-2)$ be a point interior to the region of the parabola $y^2 = 2x$ bounded by the chord joining the points (2, 2) and (8, – 4), then the number of all possible integral values of a is :

MPR188

2. The number of integral values of a for which the point $(-2a, a+1)$ will be an interior point of the smaller region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 4x$, is:

MPR189

3. A variable chord PQ of the parabola, $y^2 = 4x$ is drawn parallel to the line $y = x$. If the parameters of the points P & Q on the parabola be p & q respectively, then $(p + q)$ equal to

MPR190

4. The parabola whose axis is parallel to the y-axis and which passes through the points (0, 4), (1, 9) and (–2, 6), also passes through (2, α) then the value of α is:

MPR191

5. The centre of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point (6, 9) is (α, β) then $|\alpha - \beta|$ is

MPR192

6. If ℓ is the distance between focus and directrix of the parabola $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ then ℓ is:

Rankers Academy JEE

MPR193

7. Points A, B & C lie on the parabola $y^2 = 4ax$. The tangents to the parabola at A, B & C, taken in pairs, intersect at points P, Q & R. the ratio of the areas of the triangles PQR & ABC is

MPR194

8. A normal is drawn to a parabola $y^2 = 4ax$ at any point other than the vertex and it cuts the parabola again at a point whose distance from the vertex is not less than ka find k

MPR195

9. If three normal are drawn through $(5c, 0)$ to $y^2 = 4x$ and two of which of perpendicular then the value of c is

MPR196

10. Through the vertex O of the parabola $y^2 = 8x$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q, then the value of $\frac{OP.OQ}{10}$ is

MPR197

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y+6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is : **[JEE (Main) 2016]**

(1) $x^2 + y^2 - 4x + 9y + 18 = 0$ (2) $x^2 + y^2 - 4x + 8y + 12 = 0$

(3) $x^2 + y^2 - x + 4y - 12 = 0$ (4) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$

MPR031

2. Tangent and normal are drawn at $P(16, 16)$ on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A and B, respectively. If C is the centre of the circle through the points P, A and B and $\angle CPB = \theta$, then a value of $\tan\theta$ is- **[JEE (Main) 2018]**

(1) 2 (2) 3 (3) $\frac{4}{3}$ (4) $\frac{1}{2}$

MPR032

3. If the tangent to the parabola $y^2 = x$ at a point $(\alpha, \beta), (\beta > 0)$ is also a tangent to the ellipse, $x^2 + 2y^2 = 1$, then α is equal to : **[JEE (Main) 2019]**

(1) $2\sqrt{2} + 1$ (2) $\sqrt{2} - 1$ (3) $\sqrt{2} + 1$ (4) $2\sqrt{2} - 1$

MPR034

4. The area (in sq. units) of the smaller of the two circles that touch the parabola, $y^2 = 4x$ at the point $(1, 2)$ and the x-axis is - **[JEE (Main) 2019]**

(1) $4\pi(2 - \sqrt{2})$ (2) $8\pi(3 - 2\sqrt{2})$ (3) $4\pi(3 + \sqrt{2})$ (4) $8\pi(2 - \sqrt{2})$

MPR035

5. Let $A(4, -4)$ and $B(9, 6)$ be points on the parabola, $y^2 + 4x$. Let C be chosen on the arc AOB of the parabola, where O is the origin, such that the area of ΔACB is maximum. Then, the area (in sq. units) of ΔACB, is: **[JEE (Main) 2019]**

(1) $31\frac{3}{4}$ (2) 32 (3) $30\frac{1}{2}$ (4) $31\frac{1}{4}$

MPR037

6. Let a line $y = mx$ $(m > 0)$ intersect the parabola, $y^2 = x$ at a point P, other than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area (ΔOPQ) = 4 sq. units, then m is equal to. **[JEE (Main) 2020]**

MPR038

7. If one end of a focal chord AB of the parabola $y^2 = 8x$ is at $A\left(\frac{1}{2}, -2\right)$, then the equation of the tangent to it at B is: **[JEE (Main) 2020]**

(1) $2x + y - 24 = 0$ (2) $x - 2y + 8 = 0$

(3) $2x - y - 24 = 0$ (4) $x + 2y + 8 = 0$

MPR039

For More Join: @IITJEE_Advanced

8. Let the tangent to the parabola $S: y^2 = 2x$ at the point $P(2, 2)$ meet the x-axis at Q and normal at it meet the parabola S at the point R. Then the area (in sq. units) of the triangle PQR is equal to:

[JEE (Main) 2021]

(1) $\frac{25}{2}$ (2) $\frac{35}{2}$ (3) $\frac{15}{2}$ (4) 25

MPR042

9. Consider the parabola with vertex $\left(\frac{1}{2}, \frac{3}{4}\right)$ and the directrix $y = \frac{1}{2}$. Let P be the point where the parabola meets the line $x = -\frac{1}{2}$. If the normal to the parabola at P intersects the parabola again at the point Q, then $(PQ)^2$ is equal to: **[JEE (Main) 2021]**

(1) $\frac{75}{8}$ (2) $\frac{125}{16}$ (3) $\frac{25}{2}$ (4) $\frac{15}{2}$

MPR044

10. If a line along a chord of the circle $4x^2 + 4y^2 + 120x + 675 = 0$, passes through the point $(-30, 0)$ and is tangent to the parabola $y^2 = 30x$, then the length of this chord is: **[JEE (Main) 2021]**

(1) 5 (2) 7 (3) $5\sqrt{3}$ (4) $3\sqrt{5}$

MPR045

11. A tangent and a normal are drawn at the point $P(2, -4)$ on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If $Q(a, b)$ is a point such that $AQBP$ is a square, then $2a + b$ is equal to: **[JEE (Main) 2021]**

(1) -16 (2) -18 (3) -12 (4) -20

Rankers Academy JEE

MPR047

12. A tangent line L is drawn at the point $(2, -4)$ on the parabola $y^2 = 8x$. If the line L is also tangent to the circle $x^2 + y^2 = a$, then 'a' is equal to_____. **[JEE (Main) 2021]**

MPR048

13. The tangents at the point $A(1, 3)$ and $B(1, -1)$ on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit^2) of the triangle PAB is :- **[JEE (Main) 2022]**

(1) 4 (2) 6 (3) 7 (4) 8

MPR050

14. The equation of a common tangent to the parabolas $y = x^2$ and $y = -(x - 2)^2$ is

[JEE (Main) 2022]

(1) $y = 4(x - 2)$ (2) $y = 4(x - 1)$ (3) $y = 4(x + 1)$ (4) $y = 4(x + 2)$

MPR051

15. Let P_1 be a parabola with vertex (3, 2) and focus (4, 4) and P_2 be its mirror image with respect to the line $x + 2y = 6$. Then the directrix of P_2 is $x + 2y =$ ____. **[JEE (Main) 2022]**

MPR052

16. Let $x^2 + y^2 + Ax + By + C = 0$ be a circle passing through (0, 6) and touching the parabola $y = x^2$ at (2, 4). Then $A + C$ is equal to_____ . **[JEE (Main) 2022]**

(1) 16 (2) 88/5 (3) 72 (4) –8

MPR053

For More Join: @IITJEE_Advanced

17. Let $x = 2t,\ y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA$, where A is any point on the conic. If k is the ordinate of the centroid of ΔSAB, then $\lim\limits_{t \to 1} k$ is equal to **[JEE (Main) 2022]**

(1) $\frac{17}{18}$ (2) $\frac{19}{18}$ (3) $\frac{11}{18}$ (4) $\frac{13}{18}$

MPR054

18. Let $A(0,1), B(1, 1)$ and $C(1, 0)$ be the mid – points of the sides of a triangle with incentre at the point D. If the focus of the parabola $y^2 = 4ax$ passing through D is $\left(\alpha+\beta\sqrt{2},0\right)$, where α and β are rational numbers, then $\frac{\alpha}{\beta^2}$ is equal to **[JEE (Main) 2023]**

(1) 6 (2) 8 (3) 12 (4) $\frac{9}{2}$

MPR055

19. Let R be the focus of the parabola $y^2 = 20x$ and the line $y = mx + c$ intersect the parabola at two points P and Q. Let the point $G(10, 10)$ be the centroid of the triangle PQR. If $c - m = 6$, then $(PQ)^2$ is **[JEE (Main) 2023]**

(1) 325 (2) 317 (3) 296 (4) 346

Rankers Academy JEE

MPR057

20. Let PQ be a focal chord of the parabola $y^2 = 36x$ of length 100, making an acute angle with the positive x-axis. Let the ordinate of P be positive and M be the point on the line segment PQ such that $PM : MQ = 3 : 1$. Then which of the following points does NOT lie on the line passing through M and perpendicular to the line PQ? **[JEE (Main) 2023]**

(1) $(-3,43)$ (2) $(-6,45)$ (3) $(3,33)$ (4) $(6, 29)$

MPR058

21. If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 3, then the length of this chord is equal to _____. **[JEE (Main) 2023]**

MPR059

22. The equations of two sides of a variable triangle are $x = 0$ and $y = 3$, and its third side is a tangent to the parabola $y^2 = 6x$. The locus of its circumcentre is : **[JEE (Main) 2023]**

(1) $4y^2 - 18y - 3x - 18 = 0$ (2) $4y^2 + 18y + 3x + 18 = 0$

(3) $4y^2 - 18y + 3x + 18 = 0$ (4) $4y^2 - 18y - 3x + 18 = 0$

MPR060

23. Let the length of the focal chord PQ of the parabola $y^2 = 12x$ be 15 units. If the distance of PQ from the origin is p, then $10p^2$ is equal to ___ **[JEE (Main) 2024]**

MPR198

For More Join: @IITJEE_Advanced

24. Let a line perpendicular to the line $2x - y = 10$ touch the parabola $y^2 = 4(x - 9)$ at the point P. The distance of the point P from the centre of the circle $x^2 + y^2 - 14x - 8y + 56 = 0$ is. **[JEE (Main) 2024]**

MPR199

25. Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C? **[JEE (Main) 2024]**

(1) (2, 4) (2) (1, 2) (3) (2, 2) (4) (1, 1)

MPR200

26. Let L_1, L_2 be the lines passing through the point $P(0, 1)$ and touching the parabola $9x^2 + 12x + 18y - 14 = 0$. Let Q and R be the points on the lines L_1 and L_2 such that the ΔPQR is an isosceles triangle with base QR. If the slopes of the lines QR are m_1 and m_2. then $16\left(m_1^2 + m_2^2\right)$ is equal to _____. **[JEE (Main) 2024]**

MPR201

27. Consider the circle $C : x^2 + y^2 = 4$ and the parabola $P : y^2 = 8x$. If the set of all values of α, for which three chords of the circle C on three distinct lines passing through the point $(\alpha, 0)$ are bisected by the parabola P is the interval (p, q), then $(2q - p)^2$ is equal to ____. **[JEE (Main) 2024]**

MPR202

28. Let A, B and C be three points on the parabola $y^2 = 6x$ and let the line segment AB meet the line L through C parallel to the x-axis at the point D. Let M and N respectively be the feet of the perpendiculars from A and B on L.

Then $\left(\frac{AM \cdot BN}{CD}\right)^2$ is equal to. **[JEE (Main) 2024]**

Rankers Academy JEE

MPR203

29. Let the line $L : \sqrt{2}x + y = \alpha$ pass through the point of the intersection P(in the first quadrant) of the circle $x^2 + y^2 = 3$ and the parabola $x^2 = 2y$. Let the line L touch two circles C_1 and C_2 of equal radius $2\sqrt{3}$. If the centres Q_1 and Q_2 of the circles C_1 and C_2 lie on the y-axis, then the square of the area of the triangle PQ_1Q_2 is equal to _____. **[JEE (Main) 2024]**

MPR204

30. Let $P(\alpha,\beta)$ be a point on the parabola $y^2 = 4x$. If P also lies on the chord of the parabola $x^2 = 8y$ whose mid point is $\left(1, \frac{5}{4}\right)$. Then $(\alpha-28)(\beta-8)$ is equal to _____. **[JEE (Main) 2024]**

MPR205

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

Paragraph for Question 1 and 2

Let PQ be a focal chord of the parabolas $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a, a > 0$.

1. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then tanθ = **[JEE (Advanced) 2013]**

(A) $\frac{2}{3}\sqrt{7}$ (B) $\frac{-2}{3}\sqrt{7}$ (C) $\frac{2}{3}\sqrt{5}$ (D) $\frac{-2}{3}\sqrt{5}$

MPR079

2. Length of chord PQ is **[JEE (Advanced) 2013]**

(A) $7a$ (B) $5a$ (C) $2a$ (D) $3a$

MPR080

3. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the point P, Q and the parabola at the points R, S. Then the area of the quadrilateral $PQRS$ is – **[JEE (Advanced) 2014]**

(A) 3 (B) 6 (C) 9 (D) 15

MPR081

Paragraph for Question No. 4 and 5

Let a, r, s, t be nonzero real numbers. Let $P(at^2, 2at)$, Q, $R(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point $(2a, 0)$.

Pankers Academy JEE

4. The value of r is- **[JEE (Advanced) 2014]**

(A) $-\frac{1}{t}$ (B) $\frac{t^2+1}{t}$ (C) $\frac{1}{t}$ (D) $\frac{t^2-1}{t}$

MPR082

5. If $st = 1$, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is- **[JEE (Advanced) 2014]**

(A) $\frac{\left(t^2+1\right)^2}{2t^3}$ (B) $\frac{a\left(t^2+1\right)^2}{2t^3}$ (C) $\frac{a\left(t^2+1\right)^2}{t^3}$ (D) $\frac{a\left(t^2+2\right)^2}{t^3}$

MPR083

6. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x-3)^2 + (y+2)^2 = r^2$, then the value of r^2 is **[JEE (Advanced) 2015]**

MPR084

7. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is **[JEE (Advanced) 2015]**

MPR085

For More Join: @IITJEE_Advanced

8. Let P and Q be distinct points on the parabola $y^2 = 2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle ΔOPQ is $3\sqrt{2}$, then which of the following is(are) the coordinates of P ? **[JEE (Advanced) 2015]**

(A) $\left(4,2\sqrt{2}\right)$ (B) $\left(9,3\sqrt{2}\right)$ (C) $\left(\frac{1}{4},-\frac{1}{\sqrt{2}}\right)$ (D) $\left(1,\sqrt{2}\right)$

MPR086

9. The circle $C_1: x^2 + y^2 = 3$, with centre at O, intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3, respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centres Q_2 and Q_3, respectively. If Q_2 and Q_3 lie on the y-axis, then- **[JEE (Advanced) 2016]**

(A) $Q_2Q_3 = 12$ (B) $R_2R_3 = 4\sqrt{6}$

(C) area of the triangle OR_2R_3 is $6\sqrt{2}$ (D) area of the triangle PQ_2Q_3 is $4\sqrt{2}$

MPR087

10. Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then- **[JEE (Advanced) 2016]**

(A) $SP = 2\sqrt{5}$

(B) $SQ : QP = \left(\sqrt{5}+1\right) : 2$

(C) the x-intercept of the normal to the parabola at P is 6

(D) the slope of the tangent to the circle at Q is $\frac{1}{2}$

Rankers Academy JEE

MPR088

11. If a chord, which is not a tangent, of the parabola $y^2 = 16x$ has the equation $2x + y = p$, and midpoint (h, k), then which of the following is(are) possible value(s) of p, h and k? **[JEE (Advanced) 2017]**

(A) $p = 5, h = 4, k = -3$ (B) $p = -1, h = 1, k = -3$

(C) $p = -2, h = 2, k = -4$ (D) $p = 2, h = 3, k = -4$

MPR089

12. Let E denote the parabola $y^2 = 8x$. Let $P = (-2, 4)$, and let Q and Q' be two distinct points on E such that the lines PQ and PQ' are tangents to E. Let F be the focus of E. Then which of the following statements is (are) **TRUE**? **[JEE (Advanced) 2021]**

(A) The triangle PFQ is a right-angled triangle

(B) The triangle QPQ' is a right-angled triangle

(C) The distance between P and F is $5\sqrt{2}$

(D) F lies on the line joining Q and Q'

MPR090

Paragraph for Question No. 13 and 14

Consider the region $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ and } y^2 \leq 4 - x\}$. Let F be the family of all circles that are contained in R and have centers on the x-axis. Let C be the circle that has largest radius among the circles in F. Let (α, β) be a point where the circle C meets the curve $y^2 = 4 - x$.

13. The radius of the circle C is ____. **[JEE (Advanced) 2021]**

MPR091

For More Join: @IITJEE_Advanced

14. The value of α is ____. **[JEE (Advanced) 2021]**

MPR092

15. Consider the parabola $y^2 = 4x$. Let S be the focus of the parabola. A pair of tangents drawn to the parabola from the point $P = (-2, 1)$ meet the parabola at P_1 and P_2. Let Q_1 and Q_2 be points on the lines SP_1 and SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2. Then, which of the following is/are TRUE? **[JEE (Advanced) 2022]**

(A) $SQ_1 = 2$ (B) $Q_1Q_2 = \frac{3\sqrt{10}}{5}$

(C) $PQ_1 = 3$ (D) $SQ_2 = 1$

MPR093

16. Let P be a point on the parabola $y^2 = 4ax$, where $a > 0$. The normal to the parabola at P meets the x-axis at a point Q. The area of the triangle PFQ, where F is the focus of the parabola, is 120. If the slope m of the normal and a are both positive integers, then the pair (a, m) is

[JEE (Advanced) 2023]

(A) (2, 3) (B) (1, 3) (C) (2, 4) (D) (3, 4)

MPR094

17. Let A_1, B_1, C_1 be three points in the xy-plane. Suppose that the lines A_1C_1 and B_1C_1 are tangents to the curve $y^2 = 8x$ at A_1 and B_1, respectively. If $O = (0, 0)$ and $C_1 = (-4, 0)$, then which of the following statements is (are) TRUE? **[JEE (Advanced) 2024]**

(A) The length of the line segment OA_1 is $4\sqrt{3}$

(B) The length of the line segment A_1B_1 is 16

(C) The orthocenter of the triangle $A_1B_1C_1$ is (0, 0)

(D) The orthocenter of the triangle $A_1B_1C_1$ is (1, 0)

Rankers Academy JEE

MPR206

18. A normal with slope $\frac{1}{\sqrt{6}}$ is drawn from the point $(0, -\alpha)$ to the parabola $x^2 = -4ay$, where $a > 0$.

Let L be the line passing through $(0, -\alpha)$ and parallel to the directrix of the parabola.

Suppose that L intersects the parabola at two points A and B. Let r denote the length of the latus rectum and s denote the square of the length of the line segment AB. If $r : s = 1 : 16$, then the value of $24a$ is ___. **[JEE (Advanced) 2024]**

MPR207

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	D	B	D	C	D	D	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	B	C	C	B	A	D	D	C
Que.	21	22	23	24	25					
Ans.	B	D	C	B	D					

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B	A,D	A,B,C,D	A,B,C,D	A,B,D	A,C	A,B	A,B,C,D	A,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	A,D	A,C	B,C	A	B	C	5.00	2.00	B	
Que.	20									
Ans.	A→S; B→R; C→Q; D→P									

EXERCISE (S)

1. 1 **2.** 0 **3.** 2 **4.** 18 **5.** 23

6. 0.50 **7.** 2.00 **8.** 9.76 to 9.80 **9.** 3.00 **10.** 1.60

Rankers Academy JEE

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	3	2	4	0.50	2	1	2	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	4	2	10	1	4	2	1	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	16	3	72	10	1	68	80	36	72	192

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	D	D	B	2	4	A,D	A,B,C	A,C,D
Que.	11	12	13	14	15	16	17	18		
Ans.	D	A,B,D	1.50	2.00	B,C,D	A	A,C	12		

For More Join: @IITJEE_Advanced

Important Notes

Rankers Academy JEE

For More Join: @IITJEE_Advanced